



THE SCOTS COLLEGE

2012

HSC Pre-Trial Examination

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- Show all necessary working in Questions 8–11

Total marks – 67

Section I - Multi-choice

7 marks

- Attempt Questions 1–7
- Allow about 10 minutes for this section

Section II

60 marks

- Attempt Questions 8–11
- Allow about 1 hour 50 minutes for this section

QUESTION 5 - The length of the major axis of a hyperbola is 12. The equations of the asymptotes for the hyperbola are $y = \pm \frac{2}{3}x$. If the vertices of the hyperbola are on the y -axis, determine its equation.

A) $\frac{x^2}{16} - \frac{y^2}{36} = -1$

C) $\frac{x^2}{36} - \frac{y^2}{16} = -1$

B) $\frac{x^2}{9} - \frac{y^2}{4} = -1$

D) $\frac{x^2}{81} - \frac{y^2}{36} = -1$

QUESTION 6 - The range of the conic defined by $16x^2 + 25y^2 - 400 = 0$ is

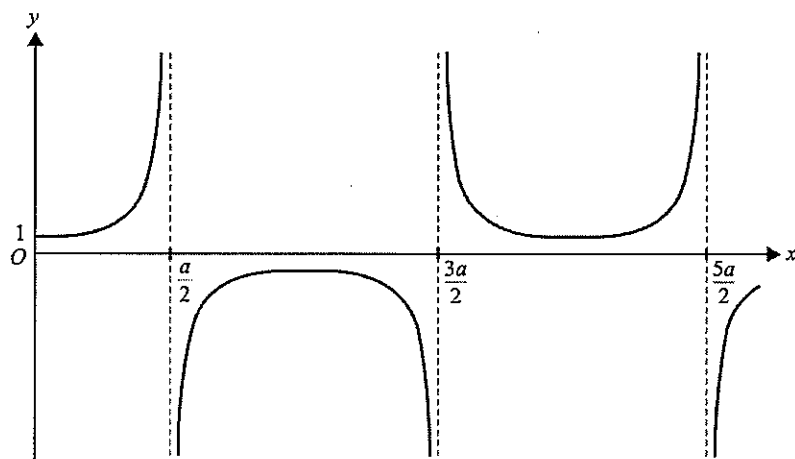
A) $|y| \leq 4$

C) $|y| < 4$

B) $|y| \leq 5$

D) $|y| < 5$

QUESTION 7



A rule for the above function is

A) $y = \operatorname{cosec}\left(\frac{2\pi}{a}\left(x - \frac{a}{2}\right)\right)$

C) $y = \operatorname{cosec}\left(\frac{\pi}{a}\left(x - \frac{a}{2}\right)\right)$

B) $y = \operatorname{cosec}\left(\frac{2\pi}{a}\left(x + \frac{a}{4}\right)\right)$

D) $y = \operatorname{cosec}\left(\frac{\pi}{a}\left(x + \frac{a}{2}\right)\right)$

Section 2 - Total marks – 60**All questions are of equal value**

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

QUESTION EIGHT (15 MARKS) Use a SEPARATE writing booklet.

a) Find $\int \frac{1}{x \log_e x} dx$ (2)

b) Using the substitution $u = e^x + 1$ or otherwise, (3)

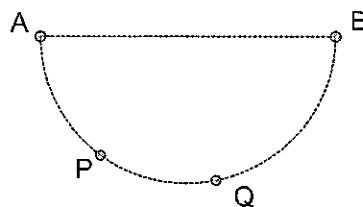
Evaluate $\int_0^1 \frac{e^x}{(1 + e^x)^2} dx$

c) i) Find a , b , and c , such that (2)

$$\frac{16}{(x^2 + 4)(2 - x)} = \frac{ax + b}{x^2 + 4} + \frac{c}{2 - x}$$

ii) Hence, or otherwise, find $\int \frac{16}{(x^2 + 4)(2 - x)} dx$ (3)

d)



Points P and Q are on the circumference of a semicircle with diameter AB .

The chords AP and BQ produced intersect at F . The chord AQ and BP intersect at G . FG produced meets AB at H .

Redraw the above diagram into your solution booklet

- i) Prove that $PFQG$ is a cyclic quadrilateral. (2)
- ii) Prove that $\angle PFG = \angle PQG = \angle PBA$. (2)
- iii) Prove that FH is perpendicular to AB . (1)

End of Question 8

QUESTION NINE (15 MARKS) Use a SEPARATE writing booklet.

a) If $z = \sqrt{3} + i$, plot on the argand diagram

i) iz (1)

ii) $z(1 + i)$ (1)

b) i) If $z = \cos \frac{\pi}{3} + i \sin \frac{\pi}{3}$, find z^5 in the domain $-\pi < \theta \leq \pi$. (2)

ii) Plot on an argand diagram, all complex numbers that are the solutions to $z^5 = 1$. (2)

c) Draw on an argand diagram, the region which satisfies both $|z - 2 + 3i| < 2$ and $\operatorname{Re}(z) \geq 3$. (2)

d) The complex numbers z_1 and z_2 are such that

$$z_1 = 5 + 2pi \quad \text{and} \quad \frac{z_1}{z_2} = 1 - i \quad \text{where } p \text{ is a real constant.}$$

i) Find z_2 in the form $a + bi$, giving the real numbers a and b in terms of p . (3)

Given that $\arg(z_2) = \tan^{-1}(4)$,

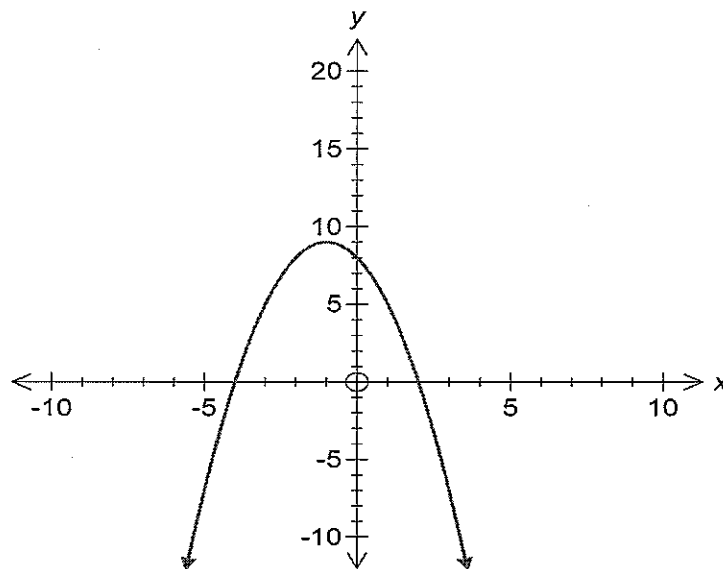
ii) Show that the value of $p = \frac{3}{2}$. (2)

iii) Find the value of $|z_1|$. (2)

End of Question 9

QUESTION TEN (15 MARKS) Use a SEPARATE writing booklet.

a)



Let $f(x) = -(x - 2)(x + 4)$, given above.

Without the use of calculus, on separate diagrams, sketch the following graphs.

i) $y^2 = f(x)$ (2)

ii) $y = \frac{1}{f(x)}$ (2)

iii) $y = e^{f(x)}$ (2)

iv) $y = \log_e(f(x))$ (2)

b) Given $y = \cos(x)$ for $0 \leq x \leq 2\pi$

i) Graph $y = |f(x)|$ (2)

ii) Where is the graph of $y = |f(x)|$ not differentiable, giving reasons for your answer. (2)

iii) Hence or otherwise find the area bounded by $y = |f(x)|$ and the x -axis. (1)

c) Sketch the curve of

$$|x| - |y| = 2 \quad (2)$$

End of Question 10

QUESTION ELEVEN (15 MARKS) Use a SEPARATE writing booklet.

a) The ellipse D has equation $\frac{x^2}{16} + \frac{y^2}{9} = 1$ and the ellipse E has equation $\frac{x^2}{4} + \frac{y^2}{9} = 1$.

- i.* Sketch D and E on the same diagram, showing the coordinates of the points where each curve crosses the axes. (2)
- ii.* Calculate the eccentricity of ellipse D (1)
- iii.* The point S is a focus of D and the point T is a focus of E . (3)
Find the length of ST .

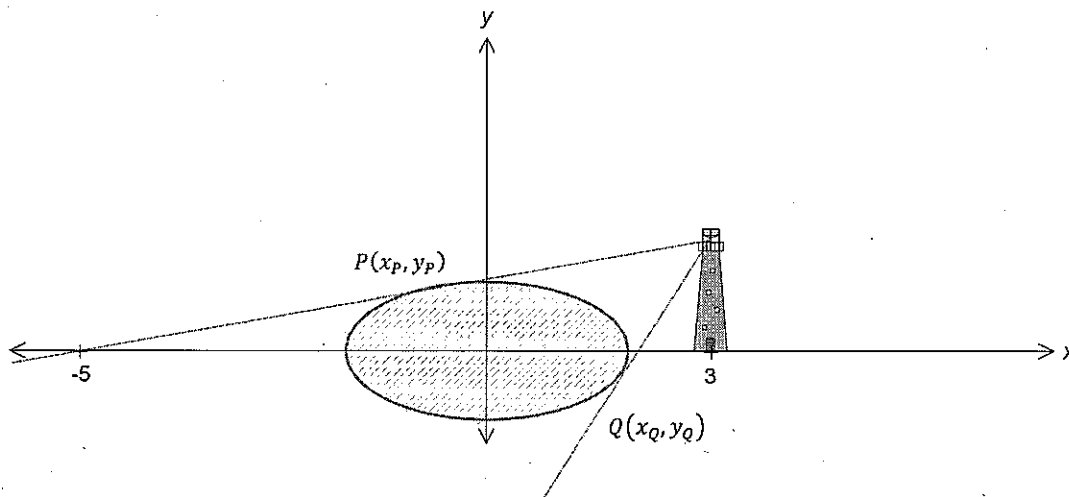
b) Determine the real values of λ for which the equation

$$\frac{x^2}{5-\lambda} + \frac{y^2}{3-\lambda} = 1 \quad \text{defines a hyperbola} \quad (1)$$

Question Eleven continues over page...

QUESTION ELEVEN - CONTINUED

c)



The diagram above shows a lighthouse, with base at $(3, 0)$, shining a light onto an elliptical shape with the equation $x^2 + 4y^2 = 5$. The rays shown above are tangents to the curve, at P and Q , and represent the boundaries of the shadow produced.

- i) Show that the equation of a tangent to the general ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at the point (x_1, y_1) is $\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$ (2)
- ii) Calculate the point of contact of the tangent at P . (1)
- iii) Hence or otherwise, show that the top of the lighthouse is located at $(3, 2)$. (1)
- iv) Hence or otherwise calculate the equation of the chord of contact of the two tangents in the above diagram. (2)
- v) Show that the angle between the two tangents is $\frac{\pi}{2} - \tan^{-1}\left(\frac{1}{4}\right) - \tan^{-1}\left(\frac{4}{11}\right)$ (2)

End of Paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Yr 12 - Ext 12 - Pretrial - solutions - 2012

Q1

$$\frac{1}{i} = \frac{i}{i^2} = -i \quad \text{(D)}$$

Q2

$$2x + 2y \frac{dy}{dx} = y + x \frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{y - 2x}{2y - x} \quad \text{(C)}$$

Q3

$$\begin{aligned} \text{let } u &= 2x+1 \\ u-1 &= 2x \\ \frac{du}{dx} &= 2 \end{aligned}$$

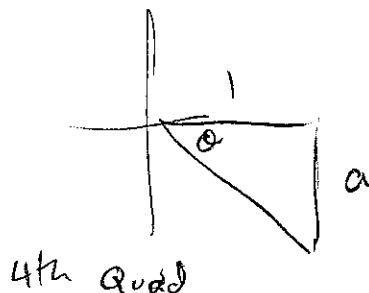
$$\frac{1}{2} du = dx$$

$$\left\{ \begin{array}{|l|} \hline x=0 \\ u=1 \\ \hline \end{array} \right\} \left\{ \begin{array}{|l|} \hline x=4 \\ u=9 \\ \hline \end{array} \right\}$$

$$\frac{1}{2} \int_1^9 (u+2)\sqrt{u} \, du$$

(D)

Q4

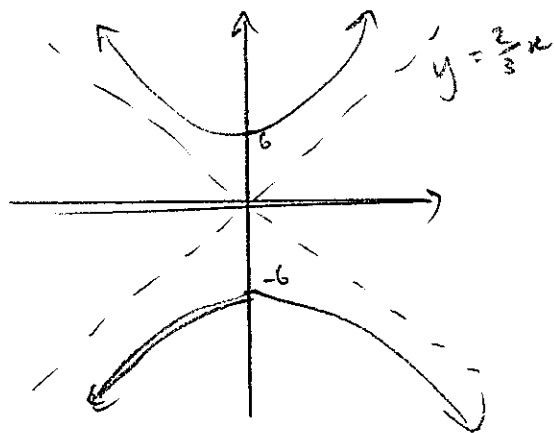


$$-\frac{\pi}{3} = \tan^{-1}\left(\frac{a}{1}\right)$$

$$\therefore a = -\sqrt{3}$$

(B)

Q5



$$y = \pm \frac{b}{a} x$$

Note: $b=6$

$$\frac{2}{3} = \frac{b}{a}$$

$$a=9$$

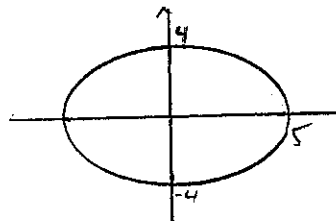
$$\frac{y^2}{36} - \frac{x^2}{81} = 1$$

$\therefore$ (D)

Q6

$$16x^2 + 25y^2 = 400$$

$$\frac{x^2}{25} + \frac{y^2}{16} = 1$$

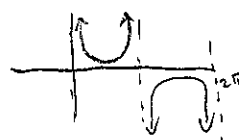
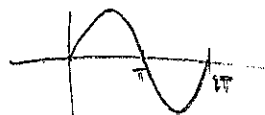


$\therefore$ (A)

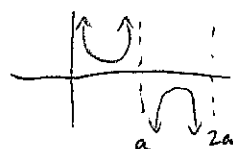
Q7

$$y = \operatorname{cosec} x = \frac{1}{\sin x}$$

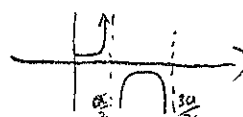
(D)



$$y = \operatorname{cosec} x$$



let $\pi = a$ $y = \operatorname{cosec}\left(\frac{\pi}{a} x\right)$



shift by $\frac{\pi}{2}$ $y = \operatorname{cosec}\left(\frac{\pi}{2} x + \frac{\pi}{2}\right)$

Q8

a) $\int \frac{1}{x} \cdot \frac{1}{\log_e x} dx$

let $u = \log_e x$

$$\frac{du}{dx} = \frac{1}{x}$$

$$du = \frac{1}{x} dx$$

$$\int \frac{1}{u} du$$

$$[\log_e(u) + C]$$

$$\log_e[\log_e x] + C$$

b) $\int_0^1 \frac{e^x}{(1+e^x)^2} dx$

$$u = e^x + 1$$

$$\frac{du}{dx} = e^x \quad \Leftrightarrow \quad du = e^x dx$$

when $\boxed{x=0}$ $\boxed{x=1}$
 $\boxed{u=e^0+1=2}$ $\boxed{u=e+1}$

$$\int_2^{e+1} \frac{1}{u^2} du = \int_2^{e+1} u^{-2} du$$

$$\left[-u^{-1} \right]_2^{e+1}$$

$$\left[-\frac{1}{e+1} + \frac{1}{2} \right]$$

Q8 6) i)
$$\frac{16}{(x^2+4)(2-x)} = \frac{(ax+b)(2-x) + c(x^2+4)}{(x^2+4)(2-x)}$$

$\therefore$

$$2ax - ax^2 + 2b - bx + cx^2 + 4c = 16$$

$\therefore$

$$c - a = 0 \Rightarrow c = a$$

$$2a - b = 0 \Rightarrow 2c - b = 0$$

$$2b + 4c = 16 \Rightarrow 4c - 2b = 0$$

Solving

$$2b + 4c = 16$$

$$-2b + 4c = 0$$

$$4b = 16$$

b	= 4
c	= 2
a	= 2

ii)

$$\int \frac{2x+4}{x^2+4} + \frac{2}{2-x} dx$$

$$\int \frac{2x}{x^2+4} + \frac{4}{x^2+4} + \frac{2}{2-x} dx$$

$$\left[\ln(x^2+4) + 2\tan^{-1}\left(\frac{x}{2}\right) - 2\ln|2-x| \right] + C$$

A geometric diagram illustrating properties of a sphere. The sphere has center \$G\$. Points \$A\$ and \$B\$ are on the upper circumference. Point \$H\$ is the midpoint of \$AB\$. A vertical line passes through \$H\$ and \$G\$, extending downwards to point \$F\$. Two points \$P\$ and \$Q\$ are on the lower circumference. Lines connect \$A\$ to \$P\$, \$A\$ to \$Q\$, \$B\$ to \$P\$, and \$B\$ to \$Q\$. Right angle symbols are shown at \$P\$ and \$Q\$ where the lines from \$A\$ and \$B\$ meet the circumference. A dashed line connects \$P\$ and \$Q\$. There are also circles with crosses (\$\otimes\$) located near \$H\$, \$G\$, and \$F\$.

as LGPF & LGQF are subtended from GF and equal 90°

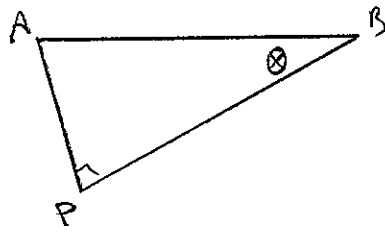
ii)

$\angle PQA = \angle PBA$ (angles subtended from same arc AP of circle to circumference are equal)

Q8d) iii)

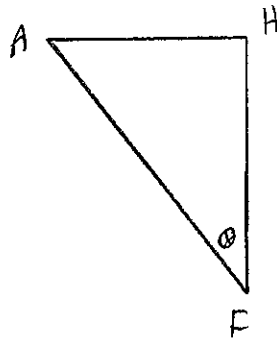
Consider

$\triangle ABP$



and

$\triangle AFH$



$$\angle ABP = \angle AFH \text{ (proven in part ii)}$$

$$\angle PAB = \angle HAF \text{ (common)}$$

$\therefore$
By angle sum of $\triangle$

$$\angle APB = \angle AHF = 90^\circ$$

$\therefore$

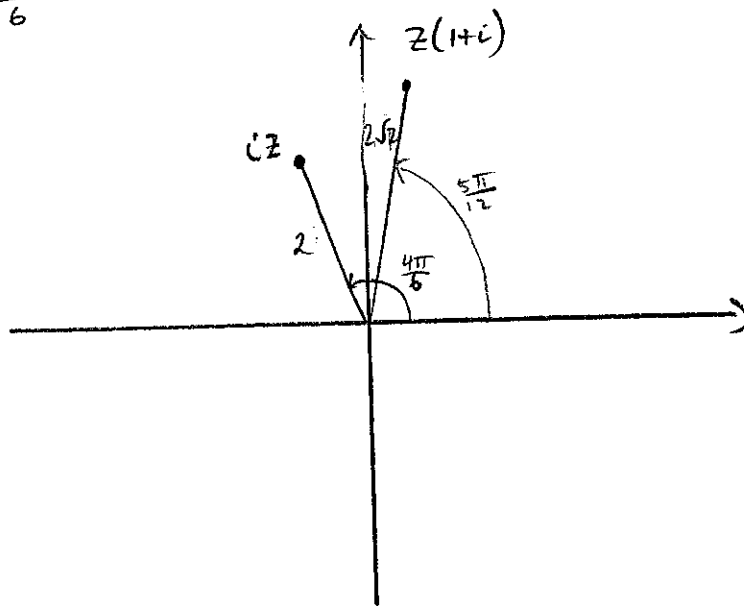
$$FH \perp AB$$

9. a)

$$z = \sqrt{3} + i$$

$$\arg(z) = \frac{\pi}{6}$$

$$|z| = 2$$



$$\begin{aligned} i) \quad iz &= 2 \cos\left(\frac{\pi}{6} + \frac{\pi}{2}\right) \\ &= 2 \cos\left(\frac{4\pi}{6}\right) \end{aligned}$$

$$\begin{aligned} ii) \quad z(1+i) &= 2\sqrt{2} \cos\left(\frac{\pi}{6} + \frac{\pi}{4}\right) \\ &= 2\sqrt{2} \cos\left(\frac{5\pi}{12}\right) \end{aligned}$$

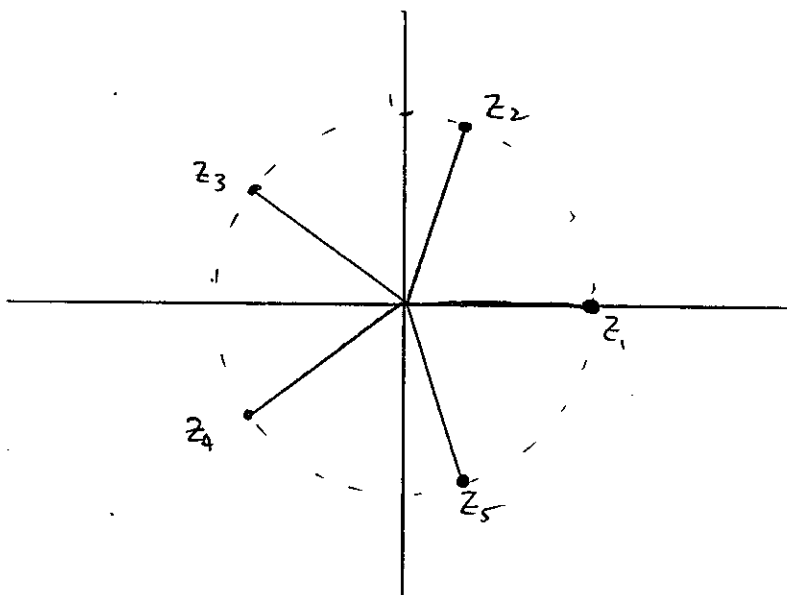
b)

$$z = \cos \frac{\pi}{3} + i \sin \frac{\pi}{3}$$

$$z^5 = \cos \frac{5\pi}{3} + i \sin \frac{5\pi}{3}$$

$$z^5 = \cos\left(-\frac{\pi}{3}\right) + i \sin\left(-\frac{\pi}{3}\right)$$

ii)



$$z_1 = 1$$

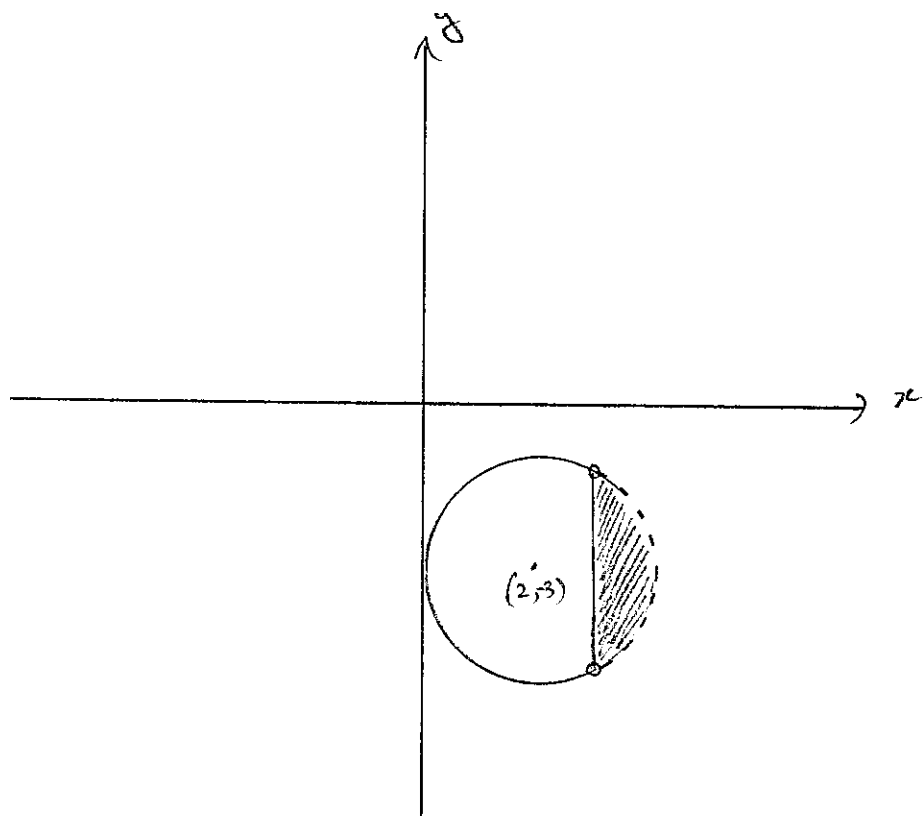
$$z_2 = \cos\left(\frac{2\pi}{5}\right)$$

$$z_3 = \cos\left(\frac{4\pi}{5}\right)$$

$$z_4 = \cos\left(-\frac{4\pi}{5}\right)$$

$$z_5 = \cos\left(-\frac{2\pi}{5}\right)$$

Q9 c)



Q9

a)

i) $z_1 = 5 + 2pi$

$$\frac{z_1}{z_2} = 1 - i$$

$\therefore$

$$z_2 = \frac{z_1}{1-i}$$

$$z_2 = \frac{5+2pi}{1-i} \times \frac{1+i}{1+i}$$

$$= \frac{5+5i+2pi-2p}{2} = \frac{5-2p}{2} + \frac{5+2p}{2}i$$

ii) $\arg(z_2) = \tan^{-1}(4)$

$$\arg(z_2) = \tan^{-1}\left(\frac{\frac{5+2p}{2}}{\frac{5-2p}{2}}\right)$$

$\therefore$

$$4 = \frac{5+2p}{5-2p}$$

$$20 - 8p = 5 + 2p$$

$$15 = 10p$$

$$\frac{15}{10} = p$$

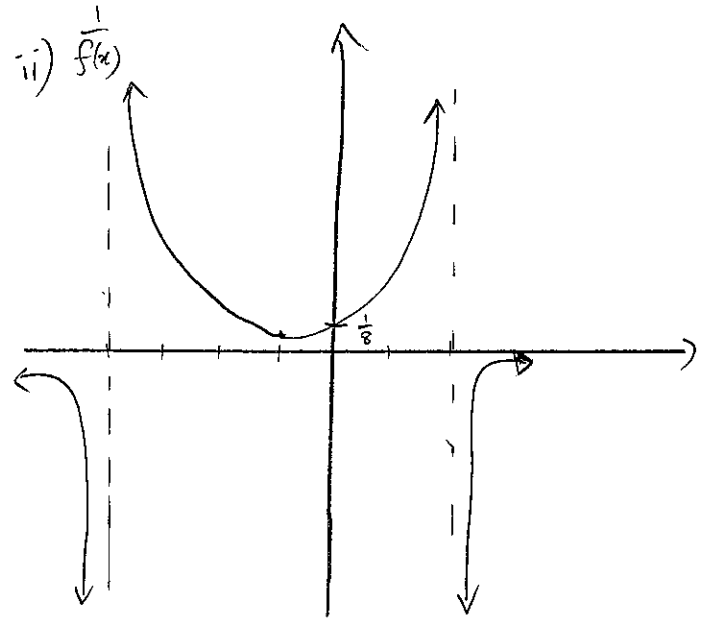
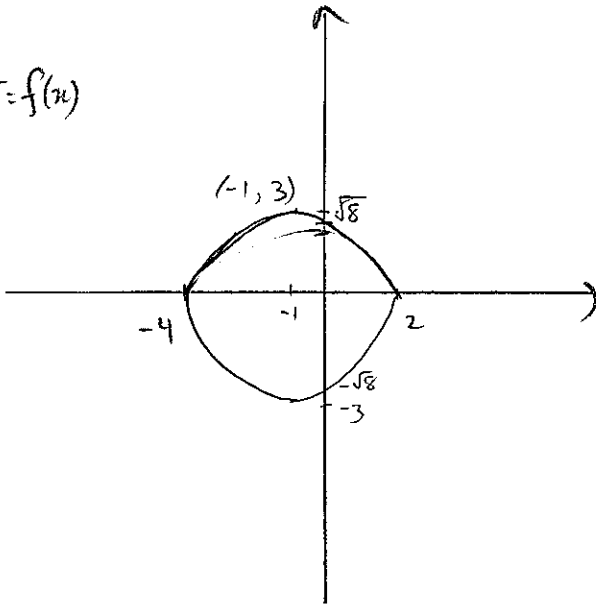
$$\frac{3}{2} = p \quad \text{as required}$$

iii) $z_1 = 5 + 3i$

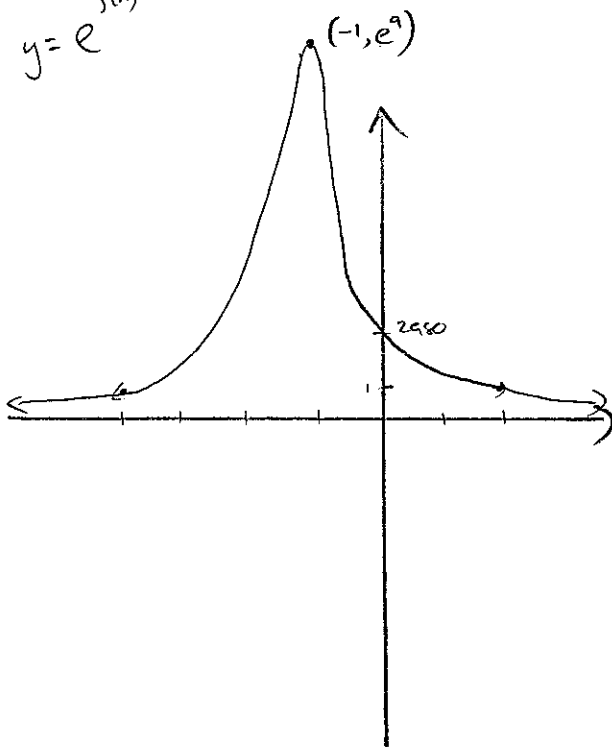
$$|z_1| = \sqrt{5^2 + 3^2}$$
$$= \sqrt{34}$$

Q10
9)

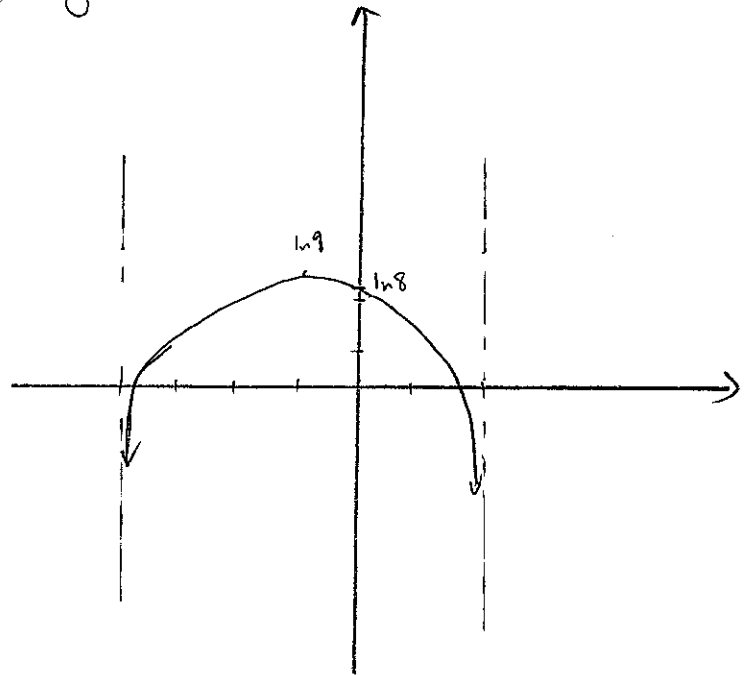
i) $y^2 = f(x)$



iii) $y = e^{f(x)}$

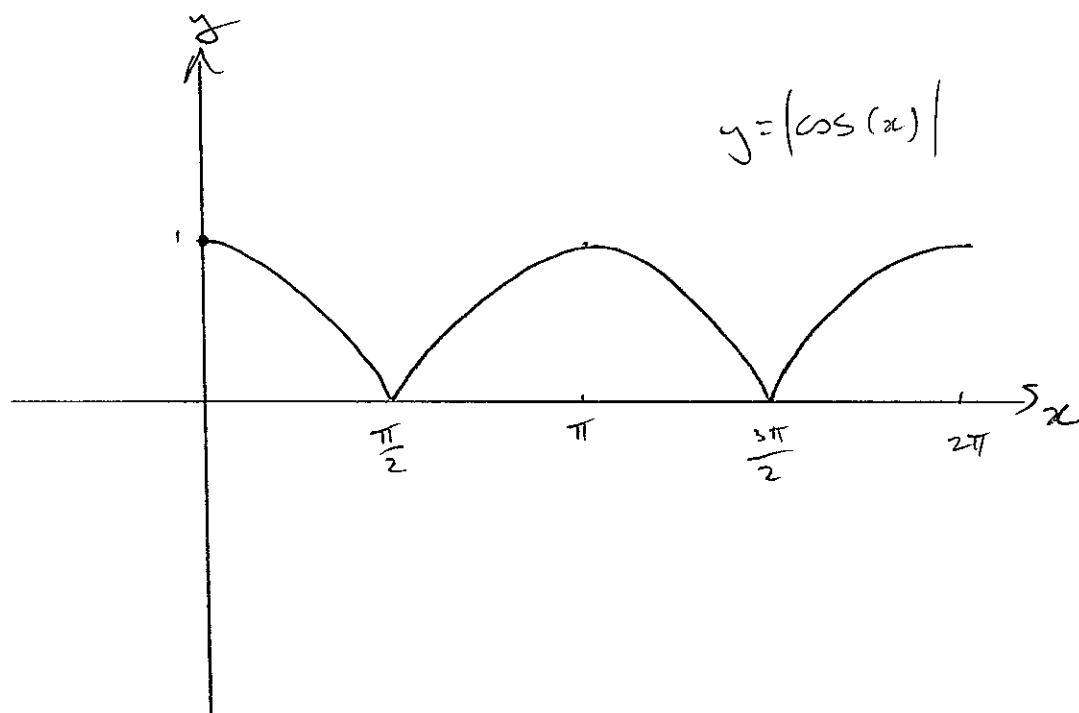


iv) $\log_e f(x)$



Q10
b)

i)



ii) Not differentiable at

$$x = \frac{\pi}{2}$$

∅

because

$$x = \frac{3\pi}{2}$$

$$\lim_{x \rightarrow \frac{\pi}{2}^+} f'(x) \neq \lim_{x \rightarrow \frac{\pi}{2}^-} f'(x)$$

$$\lim_{x \rightarrow \frac{3\pi}{2}^+} f'(x) \neq \lim_{x \rightarrow \frac{3\pi}{2}^-} f'(x)$$

Not differentiable at

$$x=0 \text{ \& } x=2\pi$$

(end points in a finite domain)

Q 10^b iii)

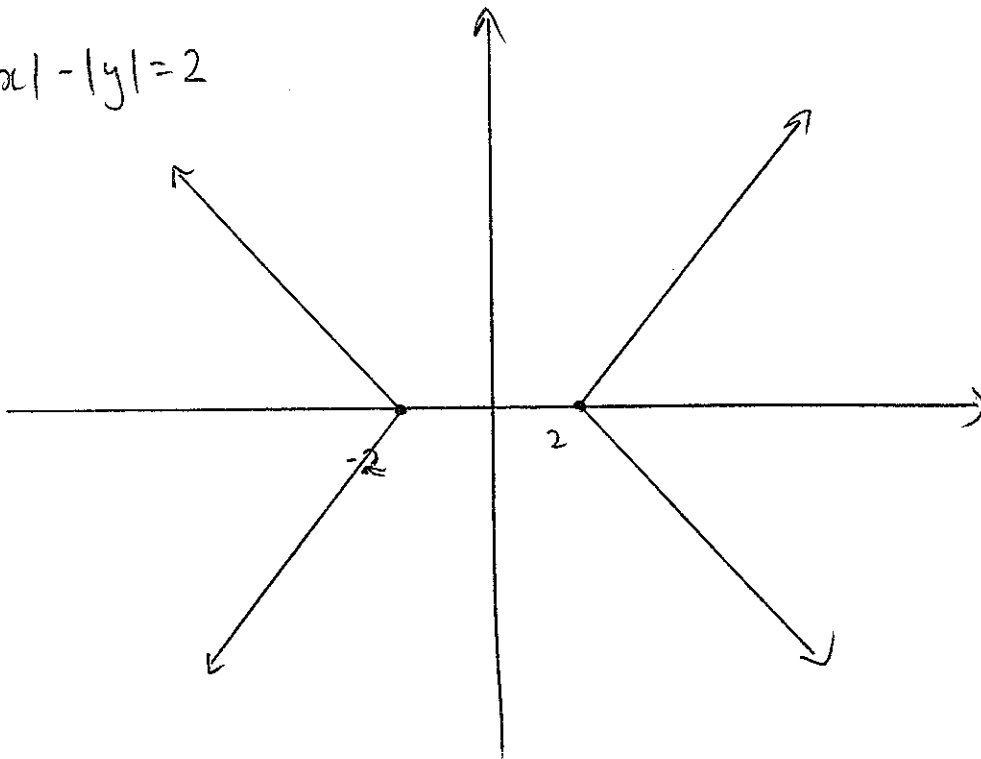
$$2 \left| \int_0^{\frac{\pi}{2}} \cos x \, dx \right|$$

$$2 \left| \left[-\sin x \right]_0^{\frac{\pi}{2}} \right| = 2 \left(0 - -\sin\left(\frac{\pi}{2}\right) \right)$$

$$= 2$$

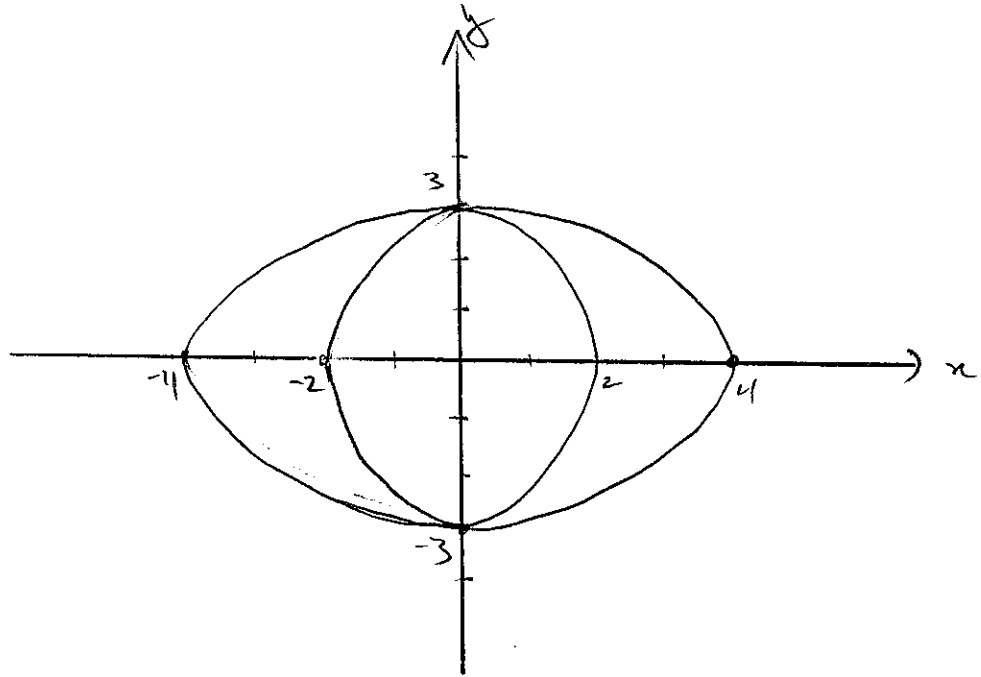
c)

$$|x| - |y| = 2$$



Q11
a)

i)



ii) $b^2 = a^2(1 - e^2)$

$$e^2 = 1 - \frac{b^2}{a^2}$$

$$e^2 = 1 - \frac{9}{16}$$

$$e = \sqrt{\frac{7}{16}} = \frac{\sqrt{7}}{4}$$

iii)

$$S = ae = \sqrt{7}$$

$$T = be = \sqrt{5}$$

$$ST = \sqrt{7 + 5}$$

$$= \sqrt{12} = 2\sqrt{3}$$

Q11 b)

$$5 - \lambda \geq 0 \quad \text{when} \quad 3 - \lambda < 0$$

$$\lambda < 5 \quad \lambda > 3$$

$$3 < \lambda < 5$$

or

$$5 - \lambda < 0 \quad \text{when} \quad 3 - \lambda \geq 0$$

$$5 < \lambda \quad \lambda \leq 3$$

$\therefore$ No solution

$\therefore$

$$3 < \lambda < 5$$

Q11 c)

i) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$$\frac{2x}{a^2} + \frac{2y}{b^2} \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{x b^2}{y a^2}$$

tangent at (x_1, y_1)

$$y - y_1 = -\frac{x_1 b^2}{y_1 a^2} (x - x_1)$$

$$2y y_1 a^2 - 2y_1^2 a^2 = -2x_1 x b^2 + 2x_1^2 b^2$$

$$\frac{y y_1}{b^2} - \frac{y_1^2}{b^2} = -\frac{x_1 x}{a^2} + \frac{x_1^2}{a^2}$$

$$\frac{x_1 x}{a^2} + \frac{y y_1}{b^2} = \frac{a_1^2}{a^2} + \frac{y_1^2}{b^2}$$

but $\frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} = 1$

$\therefore$

$$\frac{x_1 x}{a^2} + \frac{y y_1}{b^2} = 1$$

as required

Q11 c) ii) tangent at (x_p, y_p)

$$\frac{x_p x}{a^2} + \frac{y_p y}{b^2} = 1 \quad \text{satisfies } (-5, 0) \quad \text{when } a^2 = 5, b^2 = \frac{5}{4}$$

$$\frac{-5x_p}{5} + 0 = 1$$

$$-x_p = 1$$

$$x_p = -1$$

$\therefore$

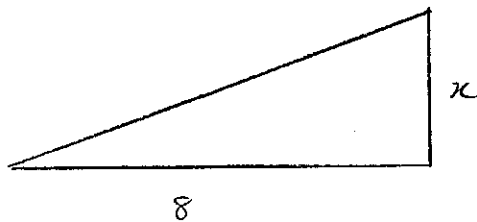
$$(-1)^2 + 4(y_p)^2 = 5$$

$$y_p = 1$$

point of contact is $(-1, 1)$

c) iii) gradient of tangent is $m = \frac{-x_p}{4y_p}$ at $(-1, 1)$

$$\therefore m = \frac{1}{4}$$



$$\frac{1}{4} = \frac{x}{8}$$

$$x = 2$$

$\therefore$ top of lighthouse is located at $(3, 2)$

Q11 c
iv) Tangent at $P(x_1, y_1)$

$$\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$$

tangent at $Q(x_2, y_2)$

$$\frac{xx_2}{a^2} + \frac{yy_2}{b^2} = 1$$

$T(x_0, y_0)$ satisfies both
such that

$\frac{xx_0}{a^2} + \frac{yy_0}{b^2} = 1$ is the chord of contact

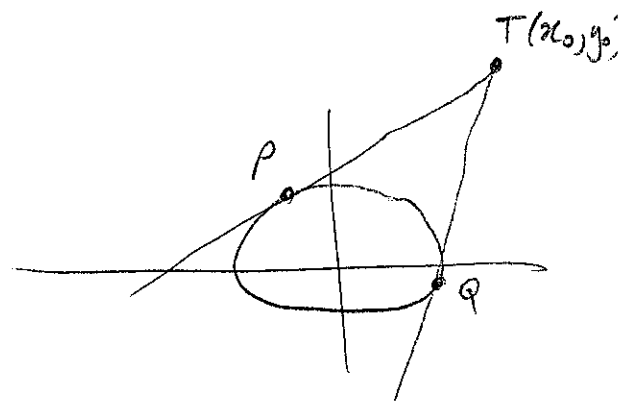
$$\text{at } (3, 2) \text{ on } \frac{x^2}{5} + \frac{y^2}{\frac{5}{4}} = 1$$

$$\frac{3x}{5} + \frac{2y}{\frac{5}{4}} = 1$$

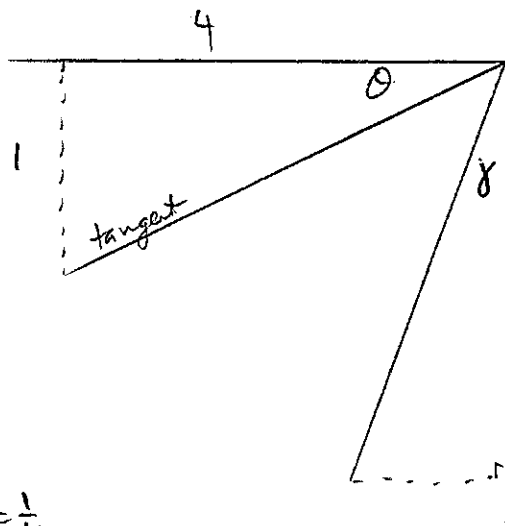
$$\frac{3x}{5} + \frac{8y}{5} = 1$$

$$3x + 8y = 5$$

$$3x + 8y - 5 = 0$$



Q11 c)
v)



Tangent at P has $m = \frac{1}{4}$

$$\therefore \tan^{-1}\left(\frac{1}{4}\right) = \theta$$

Find

$$Q(x_2, y_2)$$

$$3x + 8y = 5$$

$$y = \frac{5}{8} - \frac{3}{8}x$$

$$x^2 + 4y^2 = 5$$

$$x^2 + 4\left(\frac{5}{8} - \frac{3}{8}x\right)^2 = 5$$

$$x^2 + 4\left(\frac{25}{64} - \frac{30}{64}x + \frac{9}{64}x^2\right) = 5$$

$$25x^2 - 30x - 55 = 0$$

$$5x^2 - 6x - 11 = 0$$

$$x = \frac{6 \pm \sqrt{256}}{10}$$

$$x = \frac{-10}{10} \quad \text{or} \quad \frac{32}{10}$$

$$= -1 \quad = 3.2$$

$$Q(3.2, y_2)$$

$$3(3.2) + 8y = 5$$

$$8y = 5 - 9.6$$

$$y = -\frac{3}{16}$$

$$Q(3.2, -0.2)$$

$$M_Q = \frac{-x}{4y} = \frac{-3.2}{4(-0.2)} = \frac{11}{4}$$

$\therefore$ tangent makes angle with vertical of $\tan^{-1}\left(\frac{4}{11}\right)$

$\therefore$ Angle formed by tangents

$$\text{is } \frac{\pi}{2} - \theta - \phi$$

$$\frac{\pi}{2} - \tan^{-1}\left(\frac{1}{4}\right) - \tan^{-1}\left(\frac{4}{11}\right)$$

or

$$\tan^{-1}\left(\frac{40}{27}\right)$$